Machine learning demand forecasting and supply chain performance

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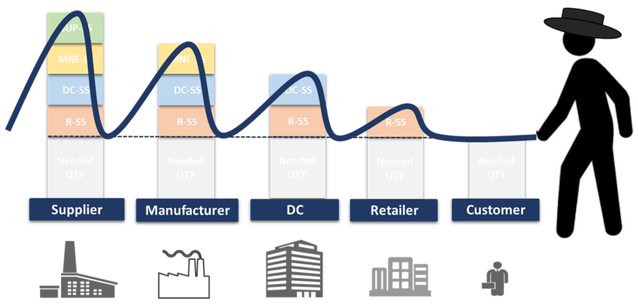
Introduction

In many supply chains, demand appears stable at the consumer level — yet grows increasingly volatile further upstream. This poster investigates why, and how machine learning can fix it.

- **Multi-echelon supply chain:** consumer → retailer → wholesaler → distributor → producer (steel products).
- **Downstream demand:** stable and forecastable close to the consumer.
- **Upstream orders:** amplified volatility — peaks and troughs disconnected from real demand.
- **Consequences:** extra costs (12.5–25%), overcapacity, excess or short stocks.
- **This phenomenon:** the **bullwhip effect**.



→ This poster shows that hybrid ML models (ARIMAX + Neural Networks + exogenous indicators) significantly outperform traditional forecasting methods, improving forecast accuracy, inventory turns, and cash-conversion cycle.

Bullwhip effect

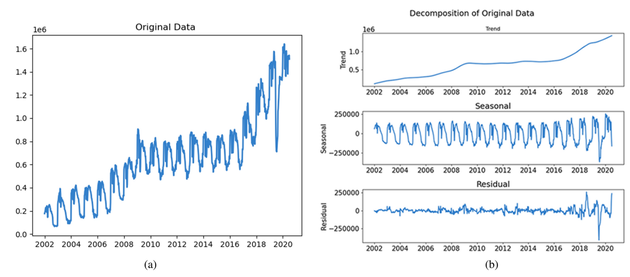
Here is the formula which describes the "Bullwhip effect"

$$D_t = d + \rho D_{t-1} + u_t$$

- D_t: the demand observed today, for example this week's sales.
- D_{t-1}: the demand from the previous period, for example last week's sales.
- d: the baseline level of demand, that is, the "normal" demand around which fluctuations occur.
- ρ: the memory effect of demand; if $\rho > 0$, a recent increase in demand tends to be followed by another increase, at least partially.
- u_t: the random noise term, that is, everything that makes demand move in an unexpected way.

Today's demand depends on an average level, a part of past demand, and an unpredictable shock.

Old methods (Simple time series)



- Simple Exponential Smoothing: used for data with no trend nor seasonality;
- Holt-Winters: Uses both the trend and the seasonality;
- Damped Trend: Used to extrapolate data assuming a linear trend that is gradually dampened (reduced) over time;
- ARIMA (Box-Jenkins); A statistical framework that identifies an "optimal" model for a series by combining Autoregressive (AR), Integrated (I), and Moving Average (MA) components.

New methods: ML-Hybrid models

- ARIMAX: Evolution of the ARIMA model. The X stands for exogenous factors that were kept to take into account not only the time serie but also causes of the change in demand. In addition to that, the predictions of the Holt-Winters model were added to make the model hybrid and exogenous;
- Neural Network with exogenous factors: The neural network is composed of 7 inputs: three months of historical demand, 3 exogenous factors (the same as for the ARIMAX model) and the forecast of the Holt-Winters model;
- Recurrent Neural Network: Specific type of neural network used with exponential smoothing. The factors of the exponential smoothing formulas were updated with the branches using the gradient;

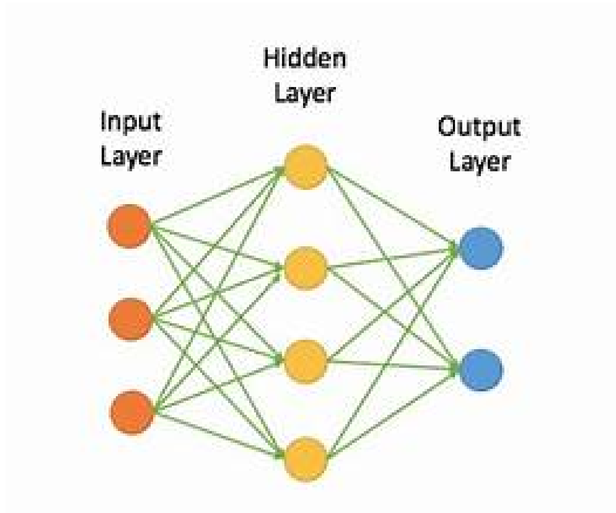
$$y_t = \alpha F_t + \sum_{i=1}^{3} \beta_i y_{t-i} + \sum_{j=1}^{3} \gamma_j w_j + e_t$$

Upsides and downsides

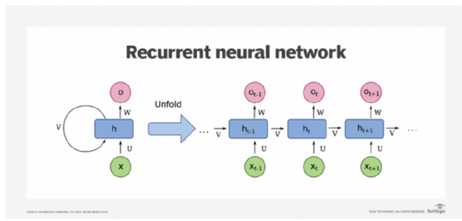
- (+) Hybrid models consistently outperform traditional methods, achieving nearly a 10% improvement in the M4 competition and a 5% accuracy gain in specific industrial applications like steel manufacturing;
- (+) These methods lead to tangible performance improvements, such as a 21.6-day reduction in the cash-conversion cycle, a 20% decrease in excess inventory, and a 15% reduction in stockouts
- (-) "Pure" machine learning methods often fail unless structurally combined with stable statistical models to handle all sources of uncertainty;
- (-) Success depends on high-quality, normalized datasets, requiring rigorous cleaning and validation to avoid unreliable predictions;

Most used method

A **neural network** is a computational model inspired by the human brain. It consists of layers of interconnected nodes called neurons. At a minimum, a network has an input layer and an output layer. In addition to that, we can add one or more hidden layers between them to complexify the function that the network can approximate. Information from the input layer propagates through the hidden layers to the output layer. Each neuron computes a weighted sum of its inputs and applies an activation function to this sum. Neural networks can learn complex patterns from data, making them very powerful tools.



Recurrent neural network (RNNs) are a class of neural networks designed to handle sequential data, where the output at each step depends on previous computations. They maintain a hidden state that captures information from prior inputs, making them effective for time series forecasting, natural language processing, and sequence prediction. Unlike feedforward networks, RNNs can model temporal dependencies. However, standard RNNs may struggle with long-term dependencies due to vanishing or exploding gradients. Advanced variants like Long Short-Term Memory (LSTM) address these limitations by improving the network's ability to learn from distant past events.



Conclusion

The bullwhip effect is a real probleme that concern a lot of different company and cost a lot of money. This effect can be reduced if multiple company in the supply chain have access to a better forecasting method. The use of new hybrid method is therefore nearly mandatory. These hybrid method are more flexible and more precise than traditionnal method or method using only machine learning. In the M4 competition 61 models were competing against each other to be the best to forecast 100.000 differents time series. The 6 first method were composed of hybrid models combining the benefit of both traditionnal model and machine learning. The machine learning alone is quite deceptive. The Neural Network method finished 60 out of 61.